I. Confidence Intervals

The people of **Statistopia** have a **population average height (μ)**, and you take **samples** to estimate it.

To get good samples, you should use:

- **Random sampling**
- **Large sample sizes**
- **Independent and identically distributed (i.i.d.) samples**

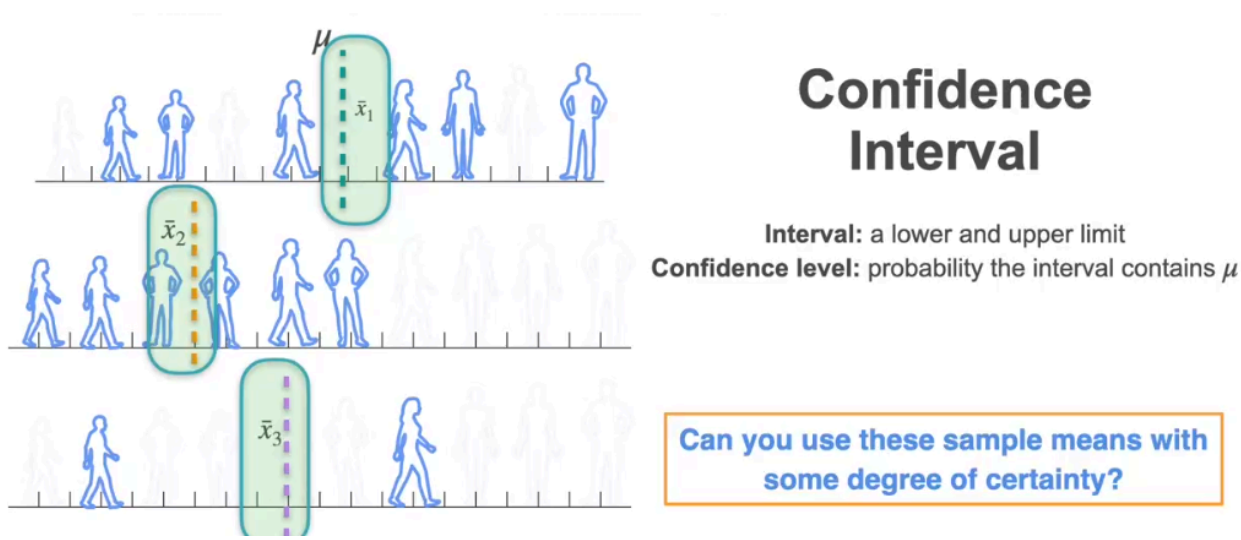
But even with these practices, no sample is **perfectly accurate**. Each time you take a sample, you get a **different sample mean**.

Since you cannot measure everyone in Statistopia to find the **true population mean**, you must rely on the **sample mean**, which always comes with **uncertainty**.

The natural question is: Can we quantify this uncertainty?

Statisticians answer this using a **Confidence Interval**.

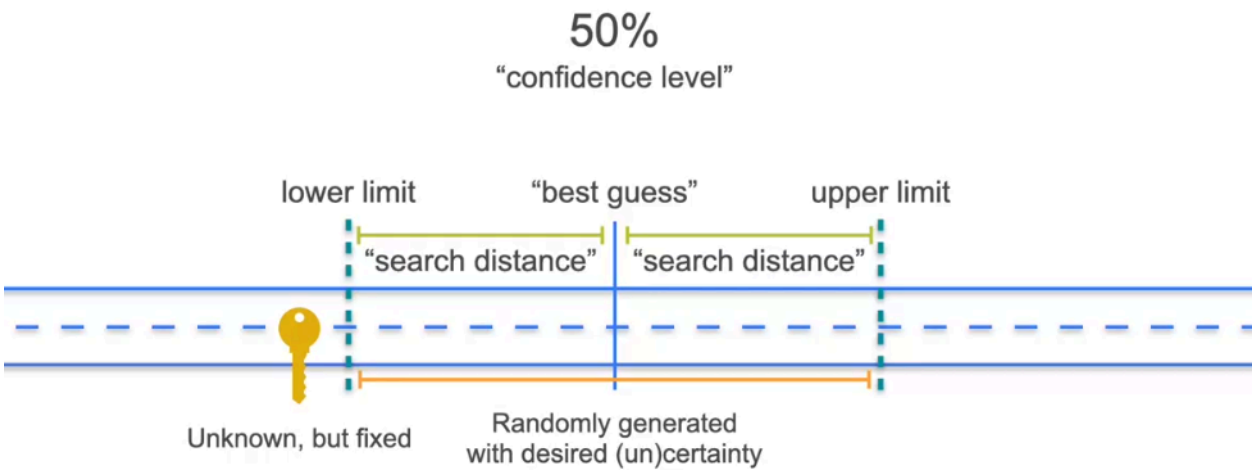
In short, a confidence interval is an interval of values that which contain the population parameter **(μ)** with some degree of certainty.



Example 1 physic's situation

Imagine you lost your key when you were heading to your friend. Your strategy to find the key is to make a **best guess** of where you dropped it, and then you and your friend search the **same distance in both directions** to cover both sides. The question is: **How large should your search distance be?** In other words, you are deciding **how confident you want to be** that you will find the key.

To define your **interval**, it's really important to remember how it is constructed. In the middle is your **best guess of the true location of the key**, and then you add a **buffer on each side** to account for the fact that you know your guess is probably wrong. The **key never moves** (just like the **true population mean** never changes), and the **orange line is the confidence interval**. The real question is: does this **confidence interval cover the population mean or not?**



The **interval** is **randomly generated** with a **desired level of certainty** — or, if you prefer, a chosen level of **uncertainty** — that it **contains the key**. I say **randomly generated** because one of the ingredients of the interval is the **best guess** you made of where to start looking.

Many people look at **confidence intervals** and say something like "50% of keys fall inside this interval," but this is **wrong**. There is only **one key**, and it has always been in the **same location**. Talking about many keys randomly appearing in different places makes no sense.

Instead, you should say: **50% of the guesses are within a search distance of the key**. This makes it clear that the **best guess** is the source of the **randomness and uncertainty**.

> **"The best guess is the source of the randomness and uncertainty, not the key itself."**

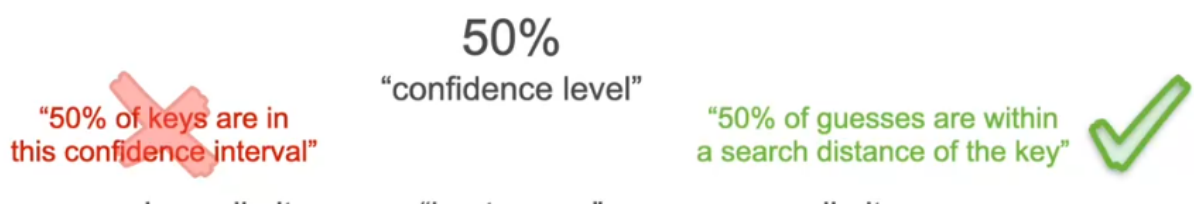
Here is what it means:

The **key (true population mean)** is **fixed**.
It does **not move**.
It is not random.

What is random is your **best guess** (the **sample mean**), because it depends on the **sample you happened to collect**. If you took a different sample, you would get a different guess.

So:

- The **true value** → fixed, not random
- The **estimate (sample mean)** → random
- The **confidence interval** → random (because it is built from the sample mean)



Example 2 : statistical situation

I.A. Estimate the mean of population

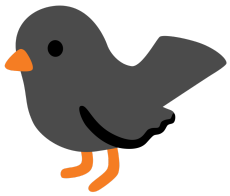
The question that we have been facing before : What is the mean population of the mean ?

The question that we have been facing before: **What does the population mean (μ)?**

For this example, the **height distribution of the population** is a **Gaussian distribution** where the **mean is unknown** but the **variance is known**. It is weird that we know the **variance** but not the **mean**, but let's take it like that just to focus on finding the **mean**. After, there is a video about finding **variance**.

Compared with the physics example, the **mean** is like the **fixed key value**. We are going to **randomly generate a confidence interval** to estimate where the **mean** is located. To do that, we are going to **sample from the population**. After, we create **$\bar{X}$ (sample mean)** and obtain a **normal distribution of sample means**.

Keep in mind:



If the **population is assumed to be Gaussian**: X~N(μ,σ2)

If you take only **one observation**, aka sample size n = 1: X=X~N(μ,σ2)

So the sampling distribution of the mean is Gaussian — but only because the original population is Gaussian.

Margin of Error and Confidence Level

To decide whether we can accept a sample mean as a good estimate of μ, we define two things:

✅ **Margin of Error** the **distance on either side of μ (or X¯)**.

✅ **Confidence Level** The **probability that the sample mean is within that margin of error**.

Before that, we define the:

✅ **Significance Level (α)**

If:

$$\alpha = 0.05$$

Then:

$$\textbf{Confidence Level} = 1 - \alpha = 0.95 = 95\%$$

This means:

> 95% of all possible randomly generated sample means fall within the central area of the sampling distribution.

And:

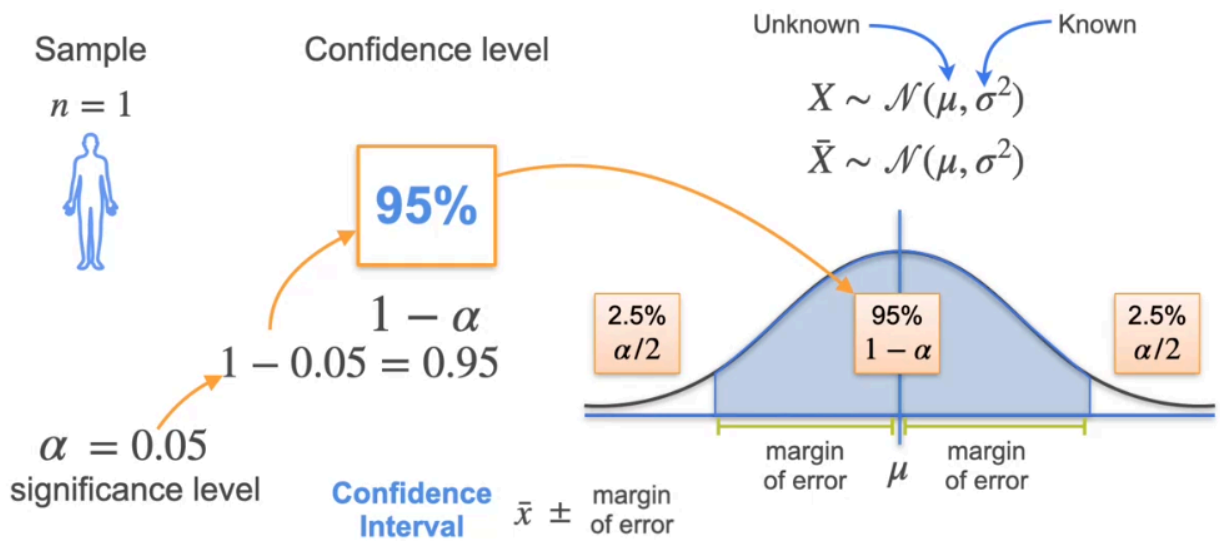
- 2.5% fall in the **right tail** (too large)
- 2.5% fall in the **left tail** (too small)

Confidence Interval Formula

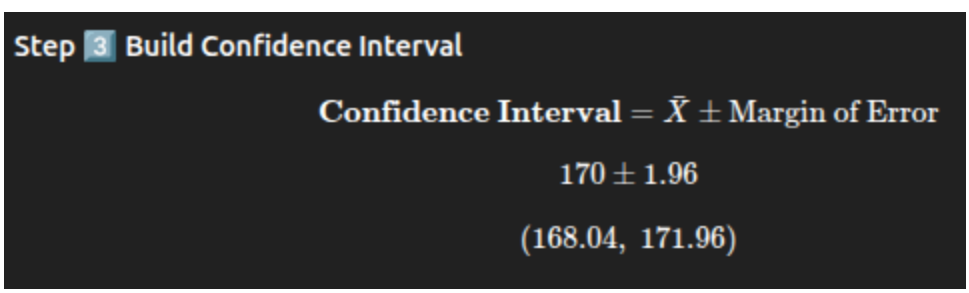
$$\text{Confidence Interval} = \bar{X} \pm \text{Margin of Error}$$

What Does This Really Mean?

If your sample mean is one of the **95% that are close to μ**, then μ must also be close to your sample mean. Unless you were very unlucky and your sample mean is unusually large or unusually small, the true mean μ should be close to your sample mean.



Practical example:



Imagine the **true population mean (μ)** is fixed somewhere on a line — but in reality, you **don't know where it is**. In the illustration, it's drawn only to help us understand what's happening.

Now you take your first sample and compute the **sample mean ($\bar{x}_1$)**. Around this value, you build a **95% confidence interval** by adding and subtracting the **margin of error**.

- In the first case, the sample mean is far from μ (maybe the person chosen is much shorter than average).
- The interval built around it does **not** contain μ.
- This interval "misses."

Then you take another sample and compute $\bar{x}_2$.

- You build a new interval.
- This time it **does contain μ**.
- This one “captures” the true mean.

You repeat again with $\bar{x}_3$. The interval again contains μ.

If you repeat this process many, many times:

- About **95% of the intervals will contain μ**
- About **5% will miss μ**

That is exactly what a **95% confidence interval** means.

It does **not** mean:

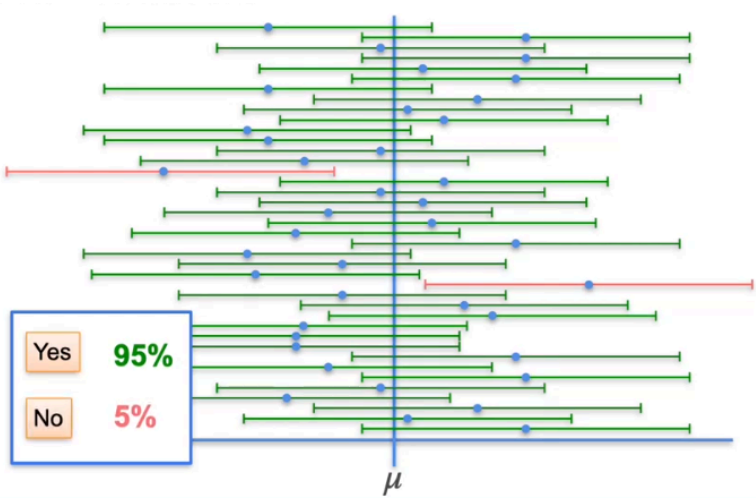
> “There is a 95% probability that μ is inside this specific interval.”

Instead, it means:

> “If we repeat this sampling method many times, 95% of the intervals we construct will contain μ.”

Important:

- **μ is fixed**
- The **interval changes** because it depends on the random sample
- Some intervals miss just by bad luck



I.1 Changing the Interval

If **n = 1**, then the **sample mean $\bar{X}$** is just the single observation itself.

Mathematically:

- When **n = 1**,

$$\bar{X} = X$$

So the **sampling distribution of the mean** is simply the **original population distribution**.

Why is it Gaussian?

In your example, the **population is assumed to be Gaussian**:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

If you take only **one observation**, then:

$$\bar{X} = X \sim \mathcal{N}(\mu, \sigma^2)$$

So the sampling distribution of the mean is Gaussian — but only because the original population is Gaussian.

What changes when n > 1?

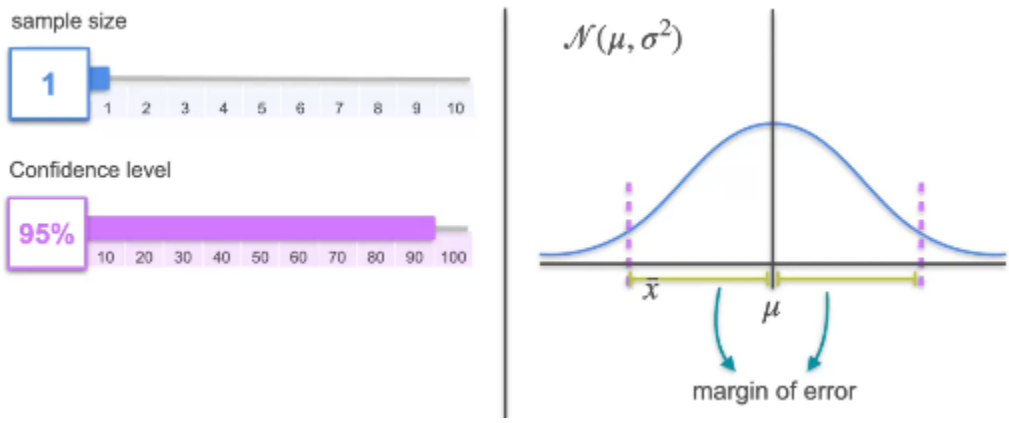
If you increase the sample size:

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

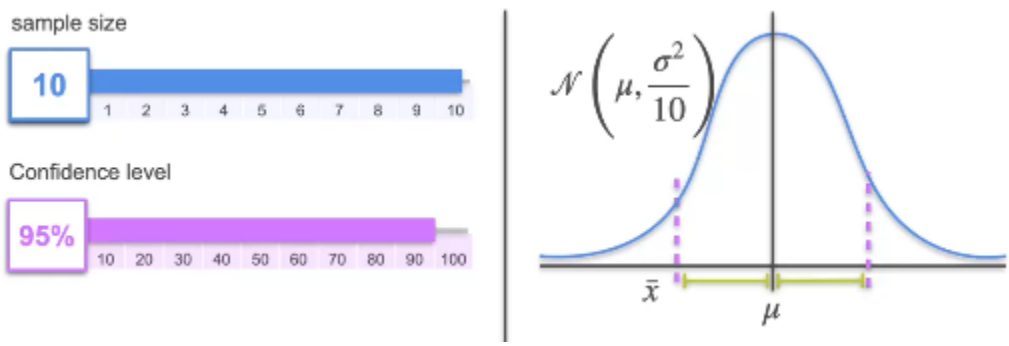
Now:

- The mean is still **μ**
- The variance becomes **σ^2/n**
- The distribution becomes **narrower**
- The estimate becomes **more precise**

Effect of sample size



Increasing size of n makes the distribution taller and narrower, also look at error margins.

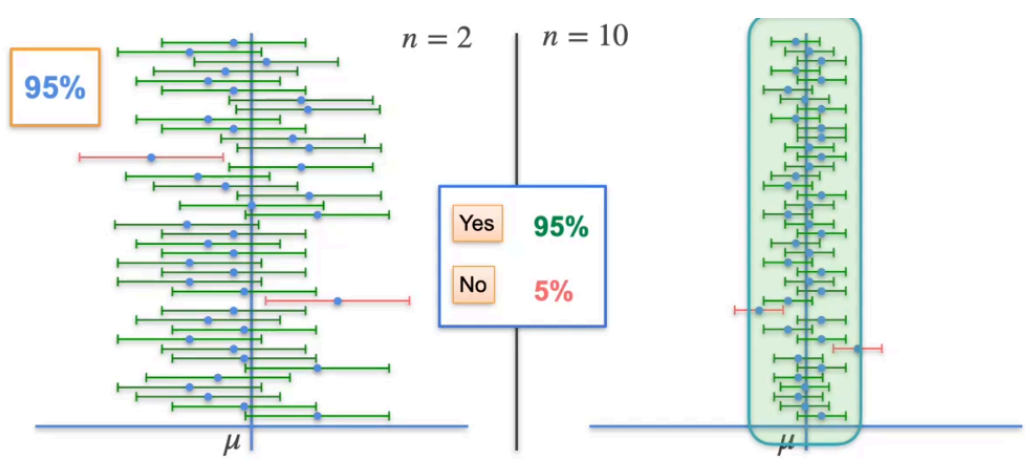


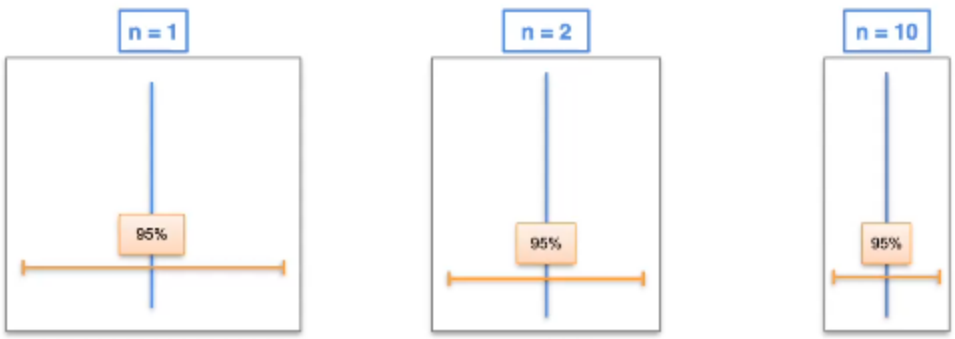
What **does change** when n increases:

- The **margin of error** decreases because the **standard error** gets smaller.
- Therefore, the **width of the confidence interval shrinks**.

So:

- **Confidence level (95%)** → stays the same.
- **Interval width** → decreases with larger n.



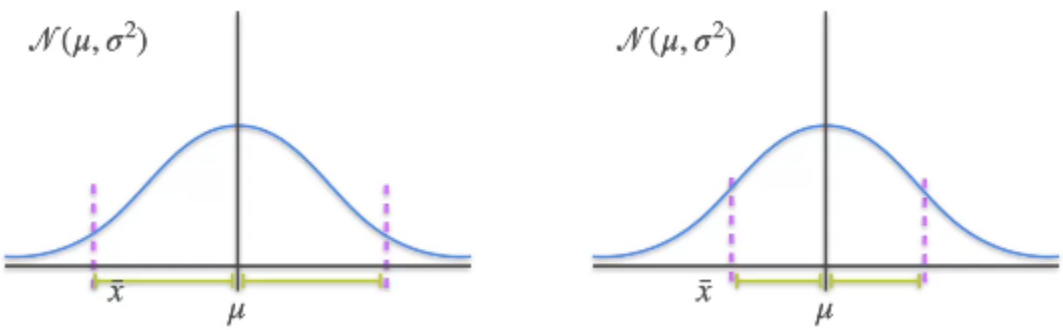


As n increases, the confidence interval shrinks

Effect of confidence Level

If we decide to **shrink the margin of error**, the **confidence level** will decrease — for example, passing from a **95% confidence level** to a **70% confidence level**.

- ☐ If you want a **higher probability** that your **interval contains the population mean**, or your **lost key**, you'll need to use **larger margins of error**.
- ☐ Higher **confidence level** → larger critical value → larger margin of error → wider interval → more likely to capture μ.



Summary

What is a confidence interval?

A range of values representing the uncertainty of the population parameter quantifying the precision of the estimate.

A confidence interval indicates the percentage of samples that contain the population parameter.

🔹 **What a Confidence Interval Is**

A **confidence interval** has two parts:

- The **sample mean** (which is random),
- A **margin of error** added and subtracted from it.

Because the **sample mean is random**, the **confidence interval is also random**.

🔹 **What the Confidence Level Means**

The **confidence level** (commonly **95%**) is the probability that intervals constructed using this method will **contain the true population mean (μ)** in repeated sampling.

- μ is **fixed but unknown**.
- You can never know for sure whether your specific interval contains μ.

🔹 Ideally, you want a **high confidence level** and a **narrow confidence interval**, but you cannot get small margins of error and high confidence for free. The most straightforward way to shrink a confidence interval is to **collect more data**, since a larger sample produces a sampling distribution with smaller variance and allows smaller margins of error.

🔹 Choosing a lower confidence level also narrows the interval, though levels below **90%** are rare and **95%** is most common. For more accurate estimates, you generally need more data.

Question

What is a confidence interval?

- (●) A range of values representing the uncertainty of the population parameter estimate and quantifying the precision of the estimate.
- ( ) A confidence interval indicates the percentage of samples that contain the population parameter.

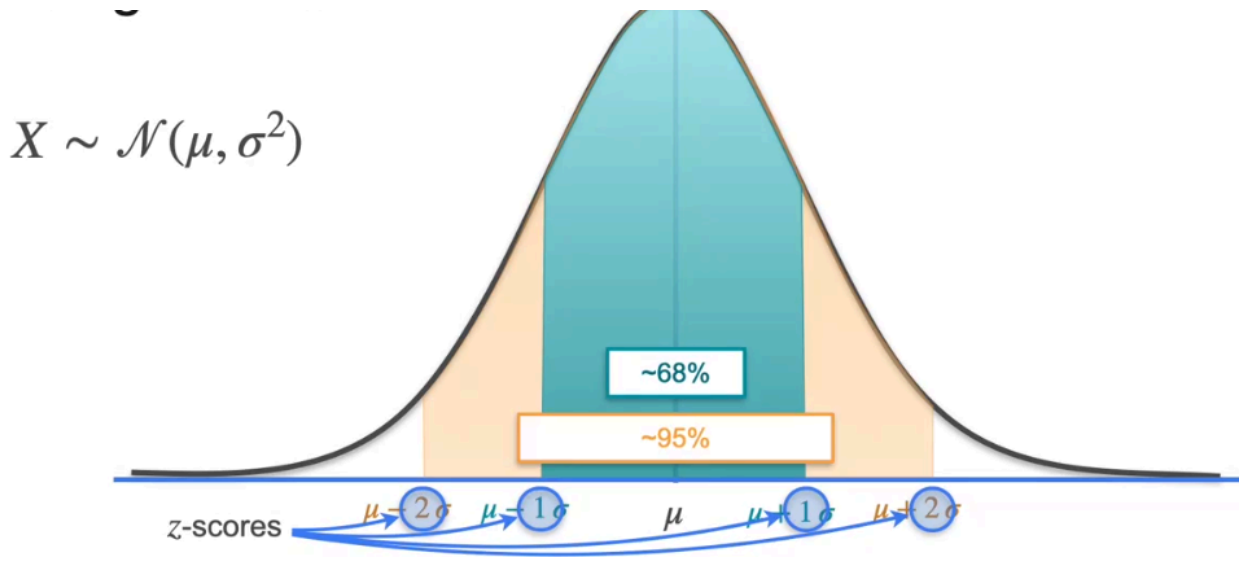
Correct
Great job!

I.2 Margin of Error

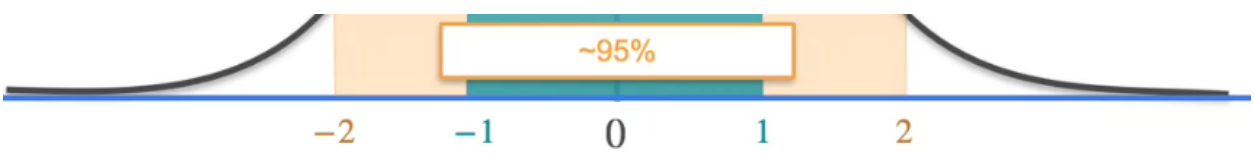
The **normal distribution** can be represented by **ranges from μ**, indicating where any **percentage of the distribution lies**.

The points away from the mean that defined the percentages have a special name called, z-score. The name z scores is taken from standard distribution z-distribution.

- The point **two standard deviations above the mean** has a **z-score of 2**.
- The point **one standard deviation below the mean** has a **z-score of −1**.



After standardization, z-score is the value of the point.



If you want exactly **95%** of the distribution, then the **z-scores** to two decimal places are **−1.96** and **+1.96**. If you randomly sample from the **standard normal distribution**, **95%** of the time the result will be inside this range. **5%** of the time your sample will fall outside this range. **−1.96** and **+1.96** are called **critical values**.

Z-score is used to know how far a value is from the mean.

$$z = \frac{x - \mu}{\sigma}$$

Also z score is used on estimating a mu of a population by controlling margin of error that will be used to generate CI for a sample mean.

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

This is used when you are looking at the **mean of a group** rather than just one person. Example: Your z-score is 2.0. This tells you that the mean of that sample is 2 standard errors above the population mean.

Those ranges change by changing alpha.

Building a Confidence Interval

A **confidence interval** has two ingredients:

- The **sample mean ($\bar{x}$)**
- The **margin of error**

Confidence interval formula:

$$\bar{x} \pm \text{margin of error}$$

Margin of Error (Known σ Case)

If the population is normal with known **μ** and **σ²**:

- The sample mean **x̄** is normally distributed
- Mean = **μ**
- Variance = **σ² / n**
- Standard deviation = **σ / √n** (called the **standard error**)

For a 95% confidence level:

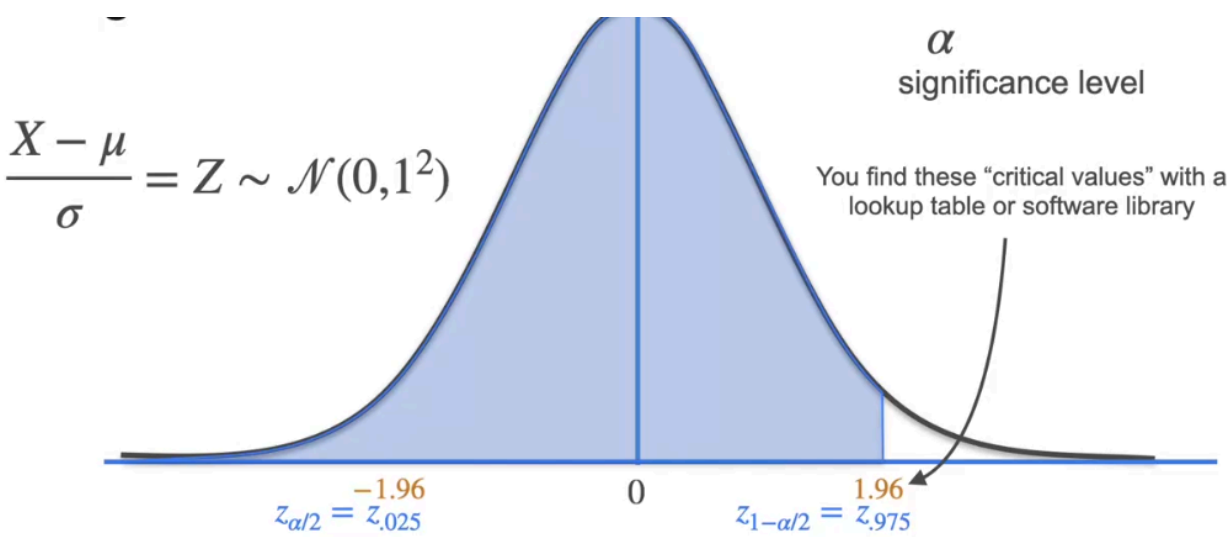
$$\text{Margin of Error} = 1.96 \times \frac{\sigma}{\sqrt{n}}$$

More generally:

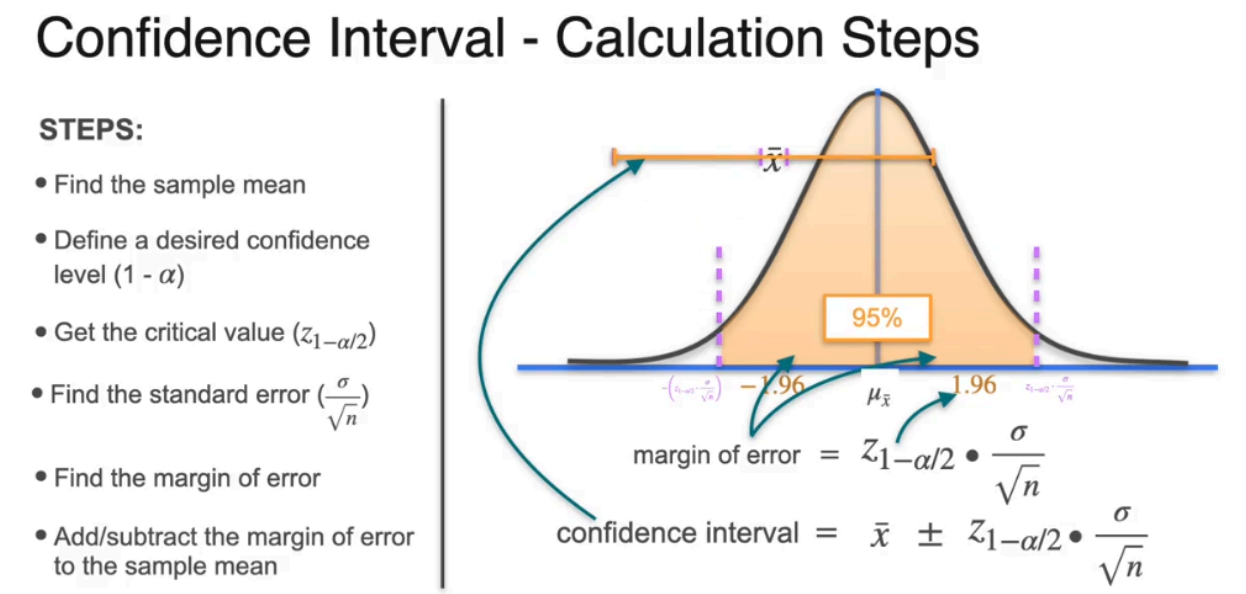
$$\text{Margin of Error} = z_{1-\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

Where:

- **z-values** are critical values from the standard normal distribution
- **α** is the significance level (e.g., 0.05 for 95%)



I.3 Confidence Intervals - Calculation Steps



Assumptions for the confidence interval to be valid:

- The **sample is random**.
- The **sample size > 30 or** the **population is approximately normal**.

Example :

Random Selection

49

$\sigma = 25cm$

95% $\rightarrow z_{1-\alpha/2} = 1.96$

Confidence Interval

$170cm \pm$ margin of error

$$\text{margin of error} = z_{1-\alpha/2} \bullet \frac{\sigma}{\sqrt{n}}$$

$$= 1.96 \bullet \frac{25}{\sqrt{49}}$$

$$= 1.96 \bullet \frac{25}{7}$$

$$= 7$$

The **confidence interval** is between **163 centimeters** and **177 centimeters**. That is the interval where we're **95% confident** that the **average height** of the people in **Statistopia** lies.

I.3 Calculating Sample Size

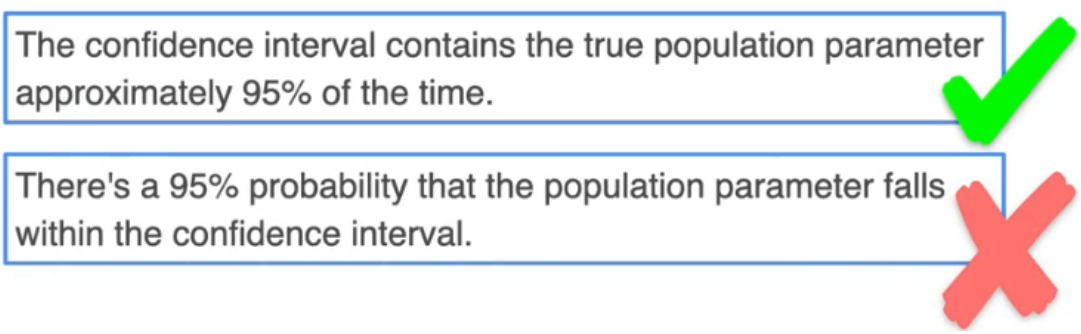
In the last example, we had a **population** of 6000 **adults**. I wanted to calculate a **95% confidence interval** for a **sample** of 49 **adults** with a **mean height** of 170, **standard deviation** of 25. We got a **margin of error** telling us that **95%** of the time the **population mean** was between **x̄ ± 7**, giving us an **interval** of 163 on the other end. But let's say that we're not happy with that **7 cm** — it is too much, we want to have a **margin of error** that is much smaller. We want, for example, **3 cm**.

What is the smallest sample size that will help us obtain the desired margin of error?

$$n \geq \left(\frac{z_{1-\alpha/2} \cdot \sigma}{\text{MOE}}\right)^2$$

$$n \geq \left(\frac{1.96 \times 25}{3}\right)^2 \approx 266.78 \approx 267$$

I.4 Difference Between Confidence and Probability



- **Correct Interpretation**
  - Saying a **95% confidence interval contains μ 95% of the time** is correct **in terms of repeated sampling**.
  - Saying there is a **95% probability that μ is in a specific interval** is **incorrect**, because μ is fixed.

- **Confidence Level**
  - Confidence level reflects the **success rate of the method**:
    - If you **repeat the sampling many times**, **95% of the intervals** calculated will contain μ.
  - It is **tied to the sample mean**, not the specific interval.

I.b. Unknown populations's Standard Deviation

Many times we don't know the variance of a population, but this is no problem, all we have to do is to make small changes by introducing t-distribution.

Recall that the **confidence interval** for a **population mean**, when we have a **point estimate** x¯bar is built using the formula:

$$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

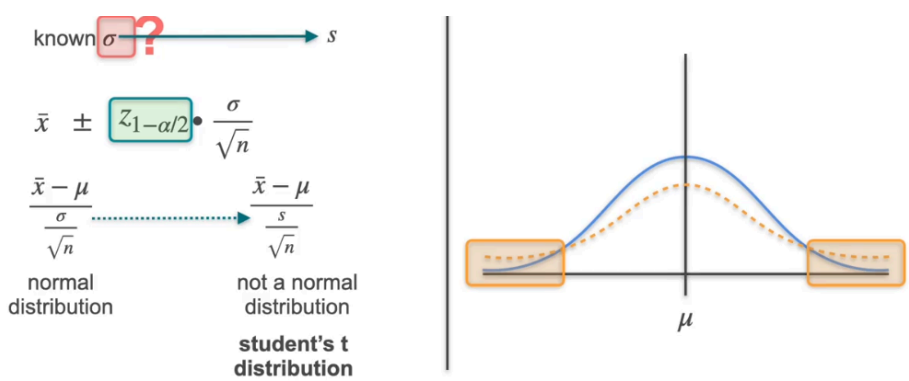
This works because we know the **population standard deviation**, σ. Knowing σ allows us to **standardize the sampling distribution** of x¯. Thanks to the **normality assumption** (or the **Central Limit Theorem** for large n), that **sampling distribution** is **normal**, centered at the true **population mean** μ, with standard deviation σ/sqr(n). Because of this **normal property**, we can use the **critical value** $z_{\alpha/2}$ from the **standard normal distribution** to capture the desired **confidence level**.

However, more often than not, we do **not know** the **standard deviation of the population**. And that's a **problem** because we can't use that σ in the formula. So what do we do in this case if we want to have a **confidence interval**? You may have guessed it — we have to use the **sample standard deviation**, s.

So we can't use the **population standard deviation**, but we took a **sample** and we can use the **sample standard deviation**, sss. This gives us this **sampling distribution** where we simply change the σ\sigmaσ into an s.

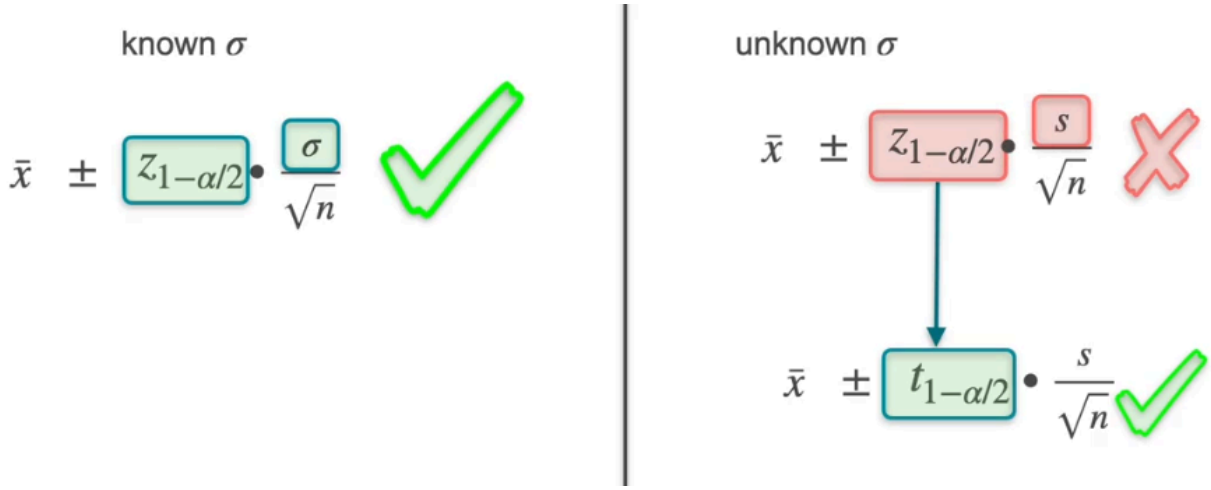
However, this quantity is **no longer a normal distribution**. Now it's a **different distribution**. It's called the **Student's t distribution**.

- **Normal population distribution** → standardization using **μ** and **σ** → **z-distribution**
- **Normal population distribution** → standardization using **x¯** and **s** → **Student's t-distribution**



T-distribution is like normal distribution but it is more fatter at the tails : meaning It has **much fatter tails**, meaning **more points** can be found **far from the center**. Sampling a point from the **t-distribution** is **more likely** to be **far from the center** than from the **normal distribution**.

To fix this **scaling issue**, we use **t-score** instead of **z-score**. To calculate **confidence intervals :**



There are several **t distributions** defined by something we call the **degree of freedom**. *

The **degree of freedom** of a **t distribution** is given by the **sample size** that we used **- 1.**

Effect of the degree of freedom in the **t distribution**: let's use a **normal distribution** as a comparison.

- The **degree of freedom** defines how **high and tight** the **t distribution** is.
- So, for a **t** with **degree of freedom 1**, you have **fatter tails**.
- Compare this to one with **5 degrees of freedom** : tailor are closer to the **normal distribution**.
- And here's one with **10 degrees of freedom, closer tails.**

The larger the number of degrees of freedom, the closer and closer you get to a normal distribution and s is more closer and closer to sigma.

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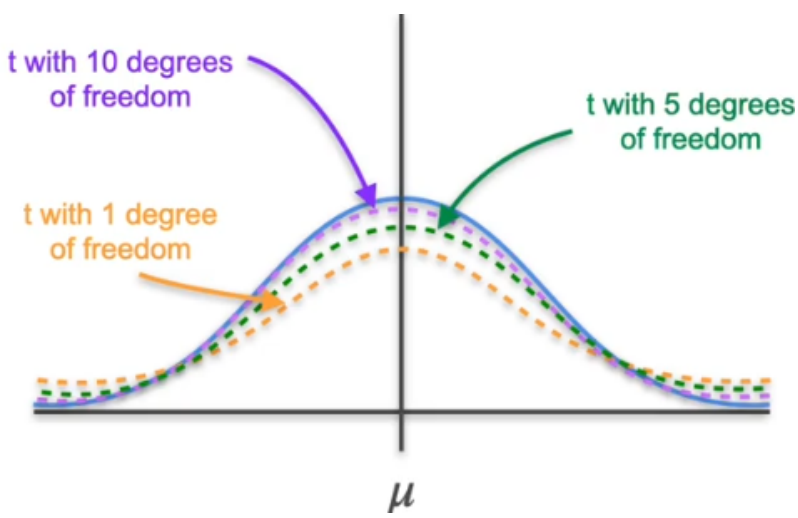


I.C. Confidence Intervals for Proportion

1. Confidence Interval for a Mean

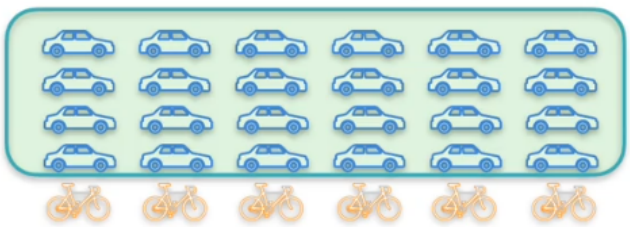
- **Data type**: **Quantitative (numeric)** data, e.g., **height, weight, blood pressure, concentration**
- **Goal**: Estimate the **average value (μ)** in a population based on a **sample**

2. Confidence Interval for a Proportion

- **Data type**: **Categorical (binary)** data, e.g., **yes/no, success/failure, infected/not infected**
- **Goal**: Estimate the **population proportion (p)** based on a **sample**

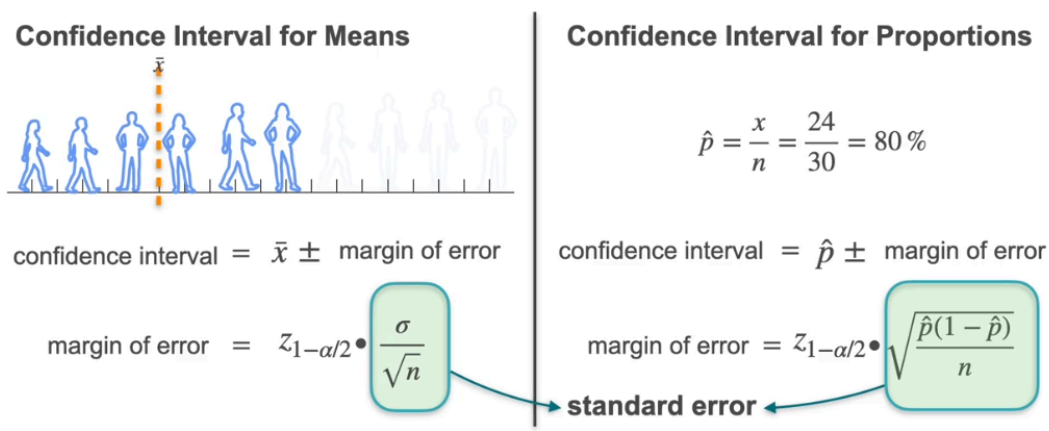
A **population proportion** is basically the **fraction of the population that has a certain characteristic**.

What is the population proportion that has a car ?



$$x = 24 \qquad \hat{p} = \frac{x}{n} = \frac{24}{30} = 80\,\%$$

$$n = 30$$



where we wanted to calculate the proportion of adults that own a car in statistopia. From the sample size of 30, we found p hat, the sample proportion to be 80%.

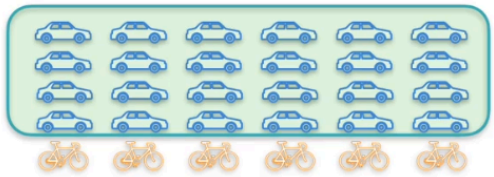
Confidence Interval for Proportions

$$\text{confidence interval} = \hat{p} \pm \text{margin of error}$$

$$\text{margin of error} = z_{1-\alpha/2} \bullet \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\text{margin of error} = 1.96 \bullet \sqrt{\frac{0.8(1-0.8)}{30}}$$

$$\text{margin of error} = 0.14$$



$n = 30 \qquad \hat{p} = 80\,\% = 0.8$

Calculate a 95% confidence interval for this sample proportion

95% $\rightarrow z_{1-\alpha/2} = 1.96$

Thus, we're 95% confident that the population proportion of how many people own a car in statistopia is within 66% and 94%.

Confidence Interval for Proportions

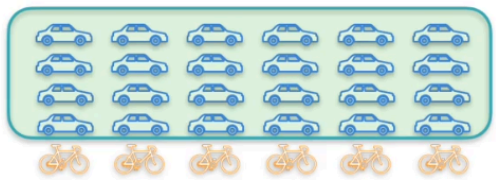
$$\text{confidence interval} = \hat{p} \pm \text{margin of error}$$

$$\text{margin of error} = 0.14$$

$$\text{confidence interval} = 0.8 \pm 0.14$$

$$0.66 < p < 0.94$$

$$66\,\% < p < 94\,\%$$



$n = 30 \qquad \hat{p} = 80\,\% = 0.8$

Calculate a 95% confidence interval for this sample proportion

95% $\rightarrow z_{1-\alpha/2} = 1.96$

II. Hypothesis testing

Hypothesis testing is a way to tell if some belief you have of the **population** is likely **True or False**.

Example of an email **detector** if it is **spam** or **ham**? In the first glance we will assume that all emails are **ham** because it is much worse if you delete a **good email** than put a **spam email** in the inbox.

So we are going to assume that all emails are good, that will be **Null Hypothesis H_0**, when we assume safely that **nothing happening**. This **null hypothesis** is called **H_0**.

And then we have an **Alternative Hypothesis H_1**. And that's the special one, that's the one we are trying to identify.

One of the important characteristics of **null hypothesis** and **alternative hypothesis** is that they are **mutually exclusive**. Because email can't be **ham** or **spam** at the same time. And accept True/False as an answer.

If there is enough **evidence** that this email is **spam**, we **reject H_0** and we **accept H_1**. However, if the gathered **evidence** is not enough to see if that email is **spam**, we **accept H_0**, but this doesn't mean the email is **ham**, only we don't have enough **evidence** to show that the email is **spam**.

- The **null hypothesis (H_0)** is the **baseline assumption** (status quo, no effect, nothing special happening).
- The **alternative hypothesis (H_1)** is the **competing statement** (there is an effect or difference).
- we usually try to find evidence for **H_1**, but we start by assuming **H_0** is true.
- The goal of **hypothesis testing** is to use **data and evidence** to decide whether to **reject H_0** or **fail to reject H_0** in favor of **H_1**.

II.1. Type I and II errors

- Type I error (sending a ham email to spam) : false positive. (an individual is good and the act is wrong).
- Type II error (identify a spam email to be ham) : false negative (an individual is bad and the act is wrong).

It is important to understand that you can **never know** if you are **right or wrong** about your decision because you do not know the **ground truth**.

However, you try your best to design a **statistical test** where you keep these **errors** below an **accepted threshold**.

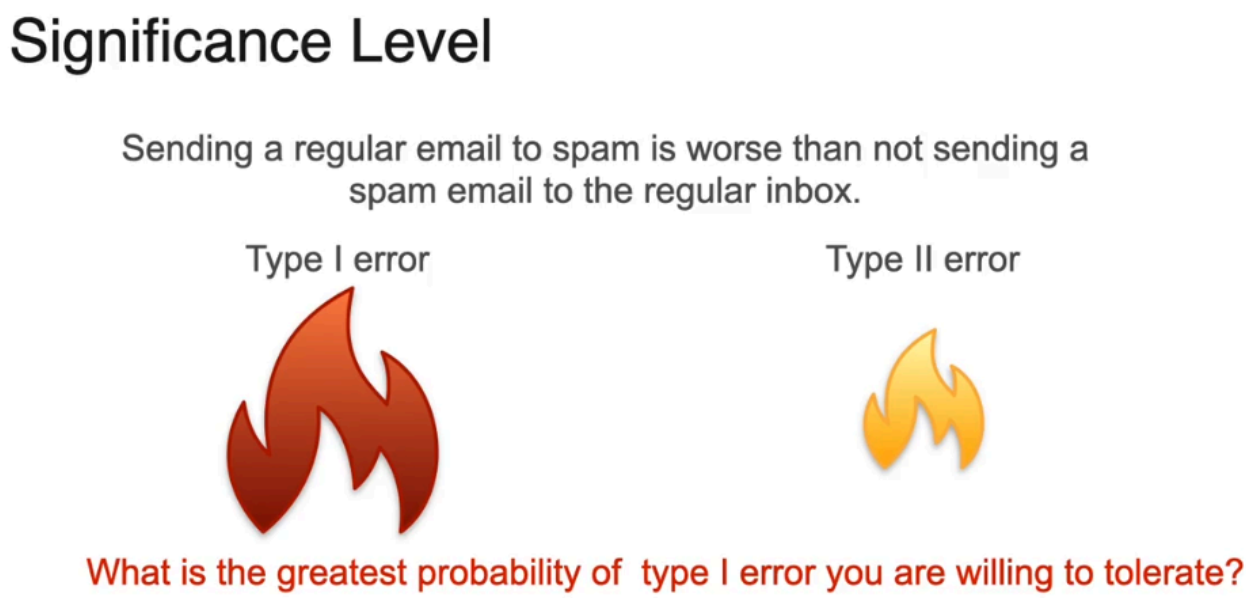
Decision	Reality	
	H_0 True (Not spam)	H_0 False (Spam)
Reject H_0 (Decide spam)	Type I error	Correct
Don't reject H_0 (Decide not spam)	Correct	Type II error

Notice that **Type I** and **Type II errors** don't have the same **impact** on the problem. In the case of the **email**, assuming that the email is **ham** implies that sending a **spam email to the inbox** is more positive than sending a **ham email to the spam box**, which is true.

You would much rather have an accidental **spam email in your inbox** than lose a perfectly good email and never be able to read it because the classifier thought it was going to be **spam**. This means that **Type I errors** are more **severe** than **Type II errors**.

The question is then how much are you willing to **negotiate** here? What is the greatest **probability of Type I error** you're willing to **tolerate**? Or in other words, on average, how many times are you willing to **incorrectly send a ham email to the spam box** in order to have a good **spam detector** that still sends most of the emails to the **correct place**?

Well, the maximum **probability of Type I error** is called the **significance level** and is usually denoted by the Greek letter **alpha (α)**. Of course, since it's a **probability**, it can only take values between **0 and 1**.



II.1.1. Significance Level

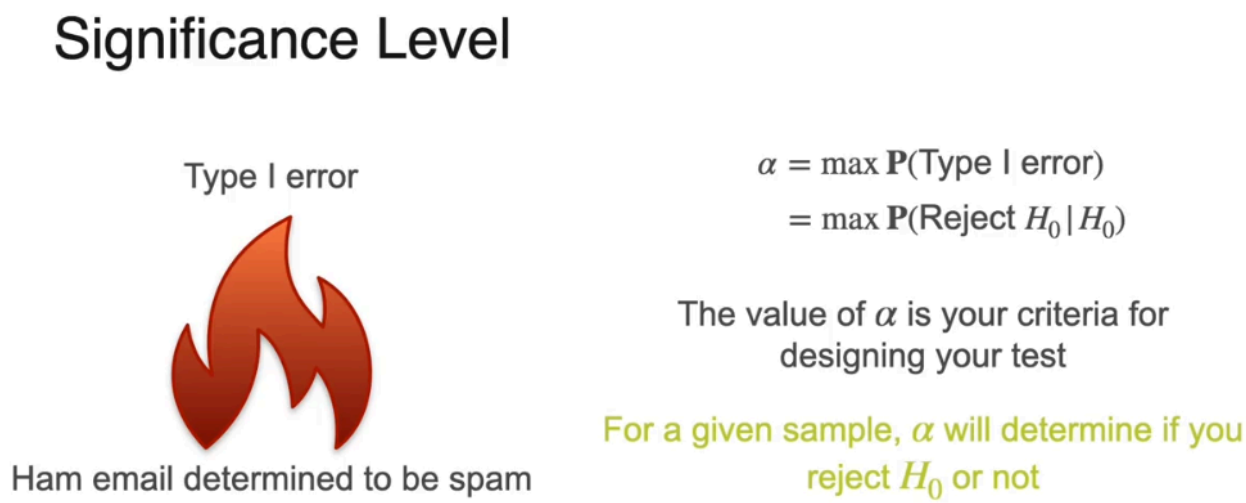
When the **significance level (α) = 0,** All emails are considered as **ham** (both **ham and spam emails** are classified as **ham**). This means you are allowing **0% probability of Type I error**. So you would **never reject H_0**. This corresponds to a **significance level (α) = 0**.

When the **significance level (α) = 1**, every email is considered as **spam**, and all of them are going to the **spam folder**. This means you allow a **100% probability of Type I error**, so you would always be willing to **reject H_0**.

Both extreme cases (**α = 0** and **α = 1**) are **terrible decision strategies**. The goal is to correctly detect **spam emails** while keeping **Type I errors** as **low as possible**. A typical **significance level (α)** is **0.05**. This means that, on average, you will incorrectly classify a **ham email as spam 5% of the time**.

- The **significance level (α)** is the **maximum probability allowed of committing a Type I error**. It is the **maximum probability of rejecting H_0 when H_0 is actually true**.

- The value of **α** is the **design criterion** of the test. **α determines the decision threshold** used to decide whether to **reject H_0 or not** based on the **sample data**.



III.2 Right-Tailed, Left-Tailed, and Two-Tailed Tests

In order to get a clear view of **hypothesis testing** and how the **significance** plays its role, let's analyze another example. Imagine you are interested in the **average height** of 18 year olds in the **US**, and consider a **sample** of the height of **10 people** measured in inches. For these observed values, the **average height** is **68.442 inches**.

Before diving into hypothesis **testing**, there's one more thing you should consider, and that is **data quality**. Since your goal is to make a decision based on **data**, then the **data** needs to be **reliable**. Otherwise, you'll be arriving at **false conclusions**.

But what does **reliable** mean? Each **sample** has to be **representative** of the **population**, the **data** needs to be completely **randomized**. This is to avoid introducing **bias** into the **decision making**. For example, if you're interested in the **height** of 18 year olds in the **US**, and you take all your **samples** from the **basketball teams** of each school, you would be introducing **bias** in the **sample** because, in general, **basketball players** are **taller**. Also, you should consider the **amount of data** you have available. Is the **sample size** enough to make a **good decision**? A good rule of thumb is considering **30 samples or more**.

Assumptions

- The **samples** are good enough for testing.
- Historical data: **mean height of 18 year olds in the US in the 1970s = 66.7 inches**.
- Observed **sample mean = 68.442 inches**.
- **Decision** will be based on **observations** (sample data), but **hypotheses** are about **population parameters**.

Hypotheses Formulation

- **Null hypothesis (H0):** population mean has not changed (**Mu = 66.7**).
- **Alternative hypothesis (H1):** population mean has changed (depends on the question).

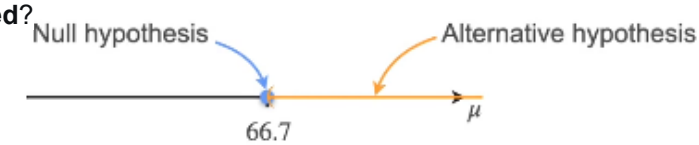
Important: Hypotheses are always about the **population**, not samples.

Test Statistics

- **Sample mean (X¯\bar{X}X¯)** is commonly used to test the **population mean**.
- **Test statistic:** a function of random samples that provides information about the **population parameter**.
    - For mean → sample mean (X¯bar)
    - For variance → s2s^2s2 or sum of squared differences from sample mean
- **Observed statistic:** the actual value from your sample (e.g., 68.442).

Types of Hypothesis Tests

1. **Right-tailed test:**
    - Question: Has the population mean **increased**?
    - H0: Mu = 66.7
    - H1: Mu > 66.7
    - Alternative is **to the right** of the null.



2. **Left-tailed test:**
    - Question: Has the population mean **decreased**?
    - H0: Mu = 66.7
    - H1: Mu < 66.7
    - Alternative is **to the left** of the null.
3. **Two-tailed test:**
    - Question: Has the population mean **changed at all**?
    - H0: Mu = 66.7
    - H1: Mu ≠ 66.7
    - Alternative extends **both left and right** of the null.

Application

Right-Tail Test (H1: Mu > 66.7)

- **Null hypothesis (H0):** Mu = 66.7
- **Alternative hypothesis (H1):** Mu > 66.7
- **Decision rule:** If the **sample mean** is much greater than 66.7, **reject H0** and accept H1.
- **Type I error:** Concluding Mu > 66.7 when **true Mu = 66.7**.
- **Type II error:** Concluding Mu = 66.7 when **true Mu > 66.7**.

Left-Tail Test (H1: Mu < 66.7)

- **Null hypothesis (H0):** Mu = 66.7
- **Alternative hypothesis (H1):** Mu < 66.7
- **Decision rule:** If the **sample mean** is much less than 66.7, **reject H0**.
- **Type I error:** Concluding Mu < 66.7 when **true Mu = 66.7**.
- **Type II error:** Not rejecting H0 when **true Mu < 66.7**.

Two-Tailed Test (H1: Mu ≠ 66.7)

- **Null hypothesis (H0):** Mu = 66.7
- **Alternative hypothesis (H1):** Mu ≠ 66.7
- **Decision rule:** Reject H0 when the **sample mean deviates significantly** from 66.7 (in either direction).
- **Type I error:** Concluding Mu ≠ 66.7 when **true Mu = 66.7**.
- **Type II error:** Not rejecting H0 when **true Mu ≠ 66.7**.

III.3 p-Value

Which of the following statements about the p-value is true?

- [x] If the p-value is less than α (significance level), then you reject the null hypothesis.

Correct
Great job! You reject the null hypothesis if the p-value is less than alpha (significance level). This decision rule is commonly used in hypothesis testing to determine whether the observed data provides enough evidence to reject the null hypothesis.

- [x] The p-value represents the probability of observing the data or more extreme results under the assumption that the null hypothesis is true.

Correct
Yes! The p-value represents the probability of observing the data or more extreme results under the assumption that the null hypothesis is true. It helps assess the strength of evidence against the null hypothesis based on the observed data.

- [x] A smaller p-value indicates stronger evidence against the null hypothesis.

Correct
Great job! A smaller p-value indicates stronger evidence against the null hypothesis. When the p-value is small, it suggests that the observed data are unlikely to have occurred if the null hypothesis were true, leading to rejection of the null hypothesis.

III.4 Critical Values
III.6 Interpreting Results

✅ Steps in Hypothesis Testing

1️⃣ State the Hypotheses

- Define the **Null Hypothesis (H_0)** → the **baseline assumption**
  - Example: H_0: $\mu = 66.7$
- Define the **Alternative Hypothesis (H_1)** → what you want to test/prove
  - Example: H_1: $\mu > 66.7$

2️⃣ Design the Test

- Choose the **test statistic** (e.g., sample mean $\bar{X}$)
- Choose the **significance level (α)**
  - Common value: **$\alpha = 0.05$**
- Remember:
  - **α = Probability of Type I error**
  - It should be **small**

3️⃣ Compute the Observed Statistic

- Calculate it using your sample data
- Example: observed mean = **68.442**

4️⃣ Make the Decision

Using the **p-value method**:

- If **p-value < α → Reject H_0**
- If **p-value ≥ α → Do not reject H_0**

(Equivalent alternative: use the **critical value method**.)

✅ Errors in Hypothesis Testing

1️⃣ Type I Error (α error)

- Happens when you **reject H_0**
- But **H_0 is actually true**
- Probability of Type I error = **Significance level (α)**
- **α is chosen by the researcher**
- α represents the **maximum probability** of making a Type I error

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2️⃣ Type II Error (β error)

- Happens when you **fail to reject H_0**
- But **H_1 is actually true**
- Probability of Type II error = **β**

⚖️ Relationship Between Errors

- For a **fixed sample size (n)**:
  - **Type I error (α)** and **Type II error (β)** are connected.
  - If you **decrease α**, you usually **increase β**.
  - If you **increase α**, you usually **decrease β**.

👉 You cannot minimize both at the same time unless you increase the **sample size**.

- *p*-values:
  - If $\mathbf{P}(\text{Reject } H_0 \,|\, H_0) < \alpha \longrightarrow$ Reject H_0 and then accept H_1
  - A small *p*-value indicates that the probability of seeing the observed data by chance is small

But is it true that **not rejecting H_0** means that the **null hypothesis is true**? This is also **false**. Remember the **spam email example**, you don't really say that the email is **ham**, the most you can guarantee is that there was **not enough evidence** to show that the email was **spam**.

- Test conclusions
  - Reject $H_0 \rightarrow H_1$ true
  - Do not reject $H_0 \rightarrow$ ~~H_0 true~~

You can only say that there is not enough evidence